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Detecting Fraud in the Graph

Graph neural networks + XGBoost, running entirely on one NVIDIA GB10

24.4 million card transactions · 589,000 scored per second · a regulator-grade explanation for every decision · no cloud, no network

MONEY20/20 MIDDLE EAST · RIYADH · 14–16 SEPTEMBER 2026

ONE MACHINE

Four financial-services demos, all on-device

Fraud detection

Graph neural network + XGBoost scoring card transactions in real time, with a Shapley explanation behind every decision.

589,528 txn/sec · F1 0.958

Portfolio optimisation

Mean-CVaR optimisation on NVIDIA cuOpt — robust tail-risk allocation fast enough to iterate on, not batch overnight.

18x vs CPU · same optimum

Document intelligence

Arabic financial document extraction into tables, text and a knowledge graph — the KYC and trade-finance back office.

on-device OCR + LLM

Loan origination

A 7B vision-language model reading identity and income documents, with a state machine driving approval and deterministic cross-document fraud checks.

Qwen2.5-VL-7B · 8 services

No cloud. No network at run time. One NVIDIA GB10 on the table.

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THE PROBLEM

One fraud in every 819 transactions

A model that says "never fraud" is 99.88% accurate — and worthless.
In the IBM TabFormer dataset: 24,386,900 transactions, 29,757 of them fraudulent. A base rate of 0.122%.

Extreme imbalance breaks conventional accuracy thinking.
The cost of a miss and the cost of a false alarm are wildly asymmetric, and both are business decisions, not model settings.

Fraud does not look unusual on its own row.
The fraudulent transactions in this dataset are mostly ordinary amounts at ordinary times. Nothing in the row screams fraud.

So the signal is not in the transaction. It is in the relationships around it.

WHAT THE DATA SAYS

The strongest fraud signals are entities, not amounts

Correlation with the fraud label, measured during preprocessing of all 24.4M rows:

Where and with whom — not when, and not how much. That is a graph problem.

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THE APPROACH

Model the transaction graph, then classify

1 Build the graph

Cardholders, transactions and merchants become nodes. Every payment becomes edges: user → transaction → merchant.

301,524 training transactions across 4,873 cardholders and 42,942 merchants.

2 Learn the neighbourhood

A GraphSAGE network aggregates each transaction's neighbourhood into an embedding — so a transaction is represented by the company it keeps, not just its own fields.

2 hops, 32 hidden channels.

3 Classify and explain

XGBoost scores the embeddings, and Shapley value sampling attributes each decision back to human-readable feature groups.

Every score arrives with its reasons.

Temporal split, not random: train before 2018, validate on 2018, test after. It is a backtest.

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THE DEMO

What the booth screen shows

Left: transactions scoring in real time. Centre: the flagged transaction and its real details. Right: why it was flagged.

THE MOMENT

An ordinary amount, flagged at 99.5%

Read the bars, not the numbers.

Merchant city dominates the decision. Postal code follows. Transaction amount contributes under 1% as much — a sliver.

A rules engine watching for large or unusual amounts never sees this transaction.

The fraud is not in the payment. It is in the company the payment keeps.

MEASURED ON THE BOX

Performance on a single NVIDIA GB10

589,528
transactions scored per second
25,803 in 0.044 s

0.958
F1 score
precision 94.3% · recall 97.3%

41.7 s
to preprocess 24.4M rows
graph construction on GPU

23.0 s
to train end to end
GNN + XGBoost

3.3 s
per live explanation
Shapley, computed on demand

9.4 s
full power-cycle recovery
container restart to serving

Portfolio optimisation: 18x vs CPU on a 51,502-variable LP. Document intelligence: full OCR, graph and vector pipeline on the same machine.

A score a bank cannot justify is a score it cannot use

Every decision carries its reasons.
Shapley attribution runs per transaction, not as a dataset-level feature-importance chart. “Why was this card declined?” has an answer.

Attribution is grouped into human language.
Not 38 anonymous encoded columns — merchant city, postal code, entry mode, terminal errors, transaction amount.

It supports the conversation with the regulator and the customer.
Adverse-action explanation, dispute handling, model risk documentation and audit all need the same thing: a defensible reason.

It also exposes the model's mistakes.
The demo displays ground truth on every transaction, including its 123 false positives and 56 misses. Confidence, not concealment.

BEYOND SUPERVISED LEARNING

A foundation model for spending

29M parameters. A 6,251-token financial vocabulary. Trained only to predict a cardholder's next transaction — never shown a fraud label.

Fraud clusters anyway.

Adding these embeddings to the production fraud features lifted average precision by 26.9% — fewer false alarms at the same catch rate.

Embeddings alone score far worse than plain features. They complement feature engineering; they do not replace it.

59,123 transactions from a 1.2M corpus · projected with cuML UMAP in 2.2 s on the GB10

Where this applies

The demo is card fraud. The architecture is a pattern for any problem where the answer lives in relationships.

USE CASE 01

Payment fraud at authorisation

Score every card transaction in the authorisation window, with the reason attached.

WHAT CHANGES	Nothing — this is the demonstrated system.
THE GRAPH	Cardholder → transaction → merchant.
WHY GRAPH WINS	Catches ordinary-looking amounts at compromised merchants that amount-and-velocity rules miss.
OPERATIONAL FIT	Sub-millisecond scoring leaves budget for the rest of the authorisation path.

STATUS Demonstrated today

USE CASE 02

Money laundering and mule networks

Find the structure, not the suspicious individual: layering chains, mule rings, circular flows.

WHAT CHANGES	Node types become accounts and counterparties; edges become transfers with amount and timing.
THE GRAPH	Account → transfer → account, across institutions where data sharing permits.
WHY GRAPH WINS	Mule networks are invisible per-account and obvious as a subgraph. This is the canonical GNN case.
OPERATIONAL FIT	Batch scoring over a transfer window; explanations support the STR narrative.

STATUS Strong architectural fit — not yet built

USE CASE 03

Merchant and acquirer risk

Score the merchant, not just the payment: onboarding risk, transaction laundering, bust-out patterns.

WHAT CHANGES	Predict on the merchant node instead of the transaction node.
THE GRAPH	Merchant ↔️ shared cardholders, terminals, beneficial owners, settlement accounts.
WHY GRAPH WINS	Fraudulent merchants are identifiable by who they share customers and infrastructure with.
OPERATIONAL FIT	Daily or weekly re-scoring; feeds portfolio monitoring and onboarding decisions.

STATUS Same node-prediction pattern

USE CASE 04

Account takeover and synthetic identity

Detect accounts that behave like a network artefact rather than a person.

WHAT CHANGES	Add device, session and identity-attribute nodes to the graph.
THE GRAPH	Customer ↔ device ↔ session ↔ shared attributes (address, phone, employer).
WHY GRAPH WINS	Synthetic identities are built from recombined real attributes — the reuse is only visible across accounts.
OPERATIONAL FIT	Real-time at login and step-up; explanations justify the friction applied.

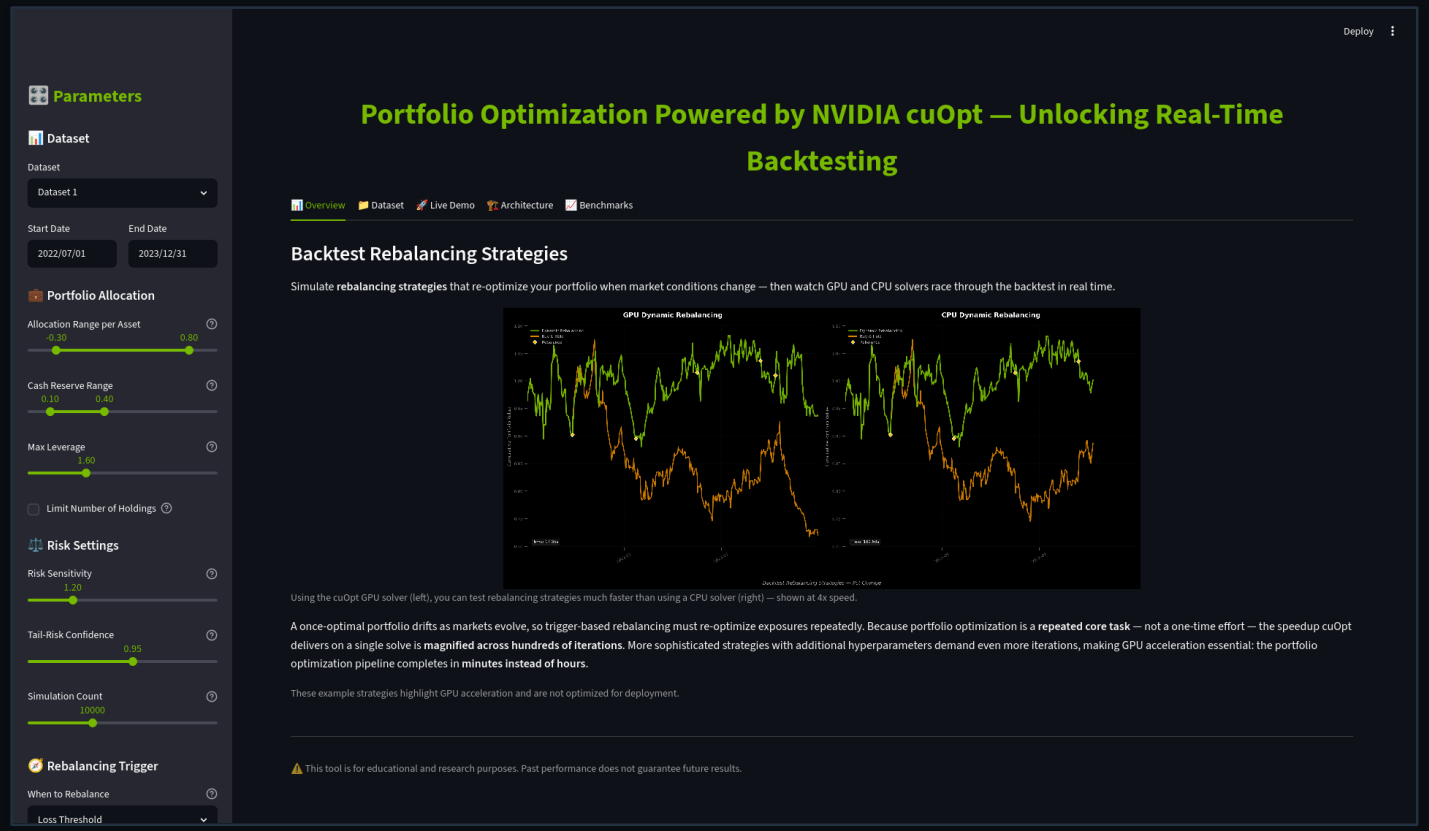
STATUS Requires additional data sources

Demo two — Portfolio optimisation

The same box, a different discipline: robust tail-risk allocation on NVIDIA cuOpt.

PORTFOLIO OPTIMISATION

Robust allocation, fast enough to iterate



Mean-CVaR asks a harder question than mean-variance: on the worst 5% of days, how much do I lose? Answering it means simulating tens of thousands of futures.

11.7 s
GPU · cuOpt

211.7 s
CPU · CVXPY

18x
faster, same optimum

500 assets × 50,000 scenarios — an LP with 51,502 variables.

Rebalancing is a repeated task, not a one-off. Across hundreds of backtest solves, that gap turns an overnight job into a morning's work.

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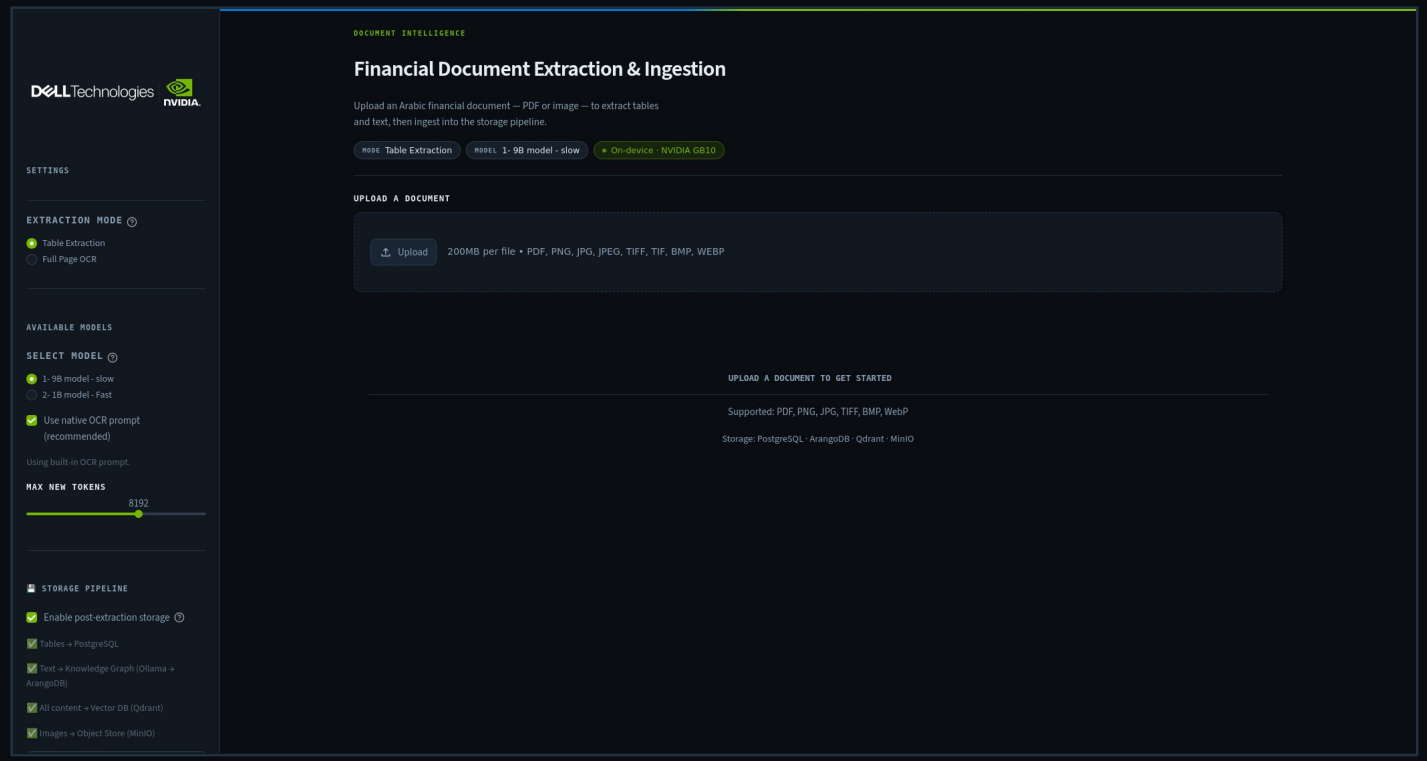
Demo three — Document intelligence

Arabic financial documents into structured data — on the same box, still offline.

Arabic documents into tables, text and a graph



KYC files, trade-finance paperwork and account-opening packs arrive as scans — often in Arabic, often as tables no rules engine can read.

The pipeline

- Layout detection and OCR on the page
- Tables → PostgreSQL, structured and queryable
- Text → knowledge graph (Ollama → ArangoDB)
- All content → vector store (Qdrant) for retrieval
- Source images → object store (MinIO)

Every model runs on the GB10. No document leaves the building.

Demo four — Loan origination

A vision-language model reading the documents a loan officer reads, and a state machine making the decision auditable.

LOAN ORIGINATION

From document upload to an auditable decision

01	**Capture**	Applicant uploads identity and income documents — national ID, employer letter, tax card, utility bill.
02	**Read**	A 7B vision-language model reads each document natively. No OCR-then-parse pipeline, no template per document type.
03	**Cross-check**	Deterministic validation across documents: does the name on the ID match the employer letter, does declared income match the tax card. Arithmetic, not inference.
04	**Decide**	A state machine walks the approval path and records why — every step reconstructable for an auditor.

Why it matters here: the fraud checks are arithmetic, not model output — so the model can be wrong without the decision being unsafe.

Data residency without a cloud dependency

The entire system is on-device.
Dataset, preprocessing, training, inference and explanation all run on a single GB10. During the demo the network cable is unplugged and nothing changes.

No transaction data leaves the institution — or the Kingdom.
That removes the cross-border data-transfer question from the procurement conversation entirely, rather than answering it.

Deployable where the data already is.
A desktop-class box fits in a branch, a data centre rack, or a regional processing site without redesigning the estate.

Recovery is measured in seconds.
Full restart to serving in 9.4 seconds, with automatic restart on power loss.

WHAT WE ARE NOT CLAIMING

The caveats, stated first

These metrics come from a rebalanced benchmark, not production conditions.

The held-out set runs at 8.09% fraud; the true base rate is 0.122%. At the real rate, false-positive economics change materially. Any pilot must be re-measured on live distributions.

Training undersamples the majority class.

A 0.1 fraud ratio with computed class weights. Standard practice for extreme imbalance, and a deliberate modelling choice rather than a hidden one.

The dataset is public and synthetic-adjacent.

IBM TabFormer, Apache 2.0. It demonstrates the method; it is not a proxy for any institution's own fraud patterns.

A production deployment needs the surrounding system.

Case management, feedback loops, model monitoring, champion/challenger and drift detection are all out of scope for this demo.

NEXT STEPS

From demo to pilot

Run it on your data.
The pipeline is three commands and roughly a minute of compute for a dataset this size. The graph schema adapts to your transaction model.

Choose the use case with the clearest data.
Payment fraud and merchant risk need only data an issuer or acquirer already holds. AML and identity graphs need more sourcing.

Measure at the true base rate from day one.
Agree the false-positive budget with the fraud operations team before tuning anything.

Deploy where the data lives.
One GB10 per site, or scale up the same containers on larger NVIDIA infrastructure — the stack is identical.

Built on the NVIDIA AI Blueprint for Financial Fraud Detection · IBM TabFormer dataset (Apache 2.0) · running on NVIDIA GB10